

SCI-NOI v0.1

SCI-NOI v0.1 r0.3 Engineering Conformance Specification

Implementation-blind Stage B draft

Scientific owner: Grant Wilson

SCI-NOI_ENGINEERING_CONFORMANCE v0.1/draft-r0.3

Not frozen. No implementation conformity, validation, calibration, physical-noise, covariance-completeness, significance, performance, readiness, or production claim.

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Document identity: SCI-NOI_ENGINEERING_CONFORMANCE v0.1/draft-r0.3

Scientific owner: Grant Wilson

Status: implementation-blind Stage B draft; not frozen.

Normative authority: the six ordered modules in SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.3, binding-file SHA-256 b7756dbeeace639e4ec86878466fb2afcc20fe5b8d256b48b6747bff1527e03c. This ECS supplies prospective evidence procedure only and authors no science.

1. Evidence identity

Every result shall identify exact product, method, parent, observation, ToITEC array, plan, generation, lifecycle, and named use. Review compares serialized scientific identities and typed states, never filenames, shapes, generic flags, approximate WCS, or undocumented defaults.

2. Ordinary operator and MAP-parent evidence

Evidence shall expose retained PTC occurrence, detector/network, coefficient family/value/QC, projection contribution, unsigned denominator, numerical coverage_cut, WCS/support, response, assignment, and application count. It shall demonstrate one sign occurrence only on Z_i^{PTC} and no changed MAP gate. A required response companion shall use identical membership and sign exactly once.

The all-+1 arithmetic shall be compared only to an independently governed, scientifically realized ordinary SCI-MAP product. Missing coefficient, coverage_cut, MAP admission, or other parent shall make m_{MAP} and dependent STD evidence unavailable.

3. Universal design evidence

Every design record shall include exact law or finite measure; stable population/order; alphabet; probabilities/weights; equality, duplicate, complement, and equivalence relations; all counts; applicable ranks/null spaces; deterministic reconstruction; lifecycle, generation, completion, and failure; and preregistration of every applicable field. Unused fields shall be not_applicable.

No review shall require every possible design to have an RNG, seed, tolerance, retry cap, or rejected-candidate counter. Those fields apply only to a selected stochastic rejection family.

4. Selected rejection-design evidence

Evidence shall enumerate the complete and active populations, zero-mass facts, zero-total not_applicable strata, unavailable ambiguous positive-mass membership, and unavailable singleton active strata. It shall reproduce exact τ_h , independent symmetric base candidates, first-accepted search, positive cap, fail-closed exhaustion, with-replacement draws, separate exact duplicate/orbit relations, equal $1/N_{\text{resolved}}$ weights, uncentered r_{sign} , and content-bound scheduling-independent generator/key identity.

An exact enumeration or analytic proof shall establish conditional uniformity: for each admitted vector x , the probability that trial k is the first accepted vector and equals x is $(1-q)^{(k-1)} p$, where common candidate probability p is identical for every admitted vector and q is total admission probability. Summing $k=1..\text{cap}$ supplies the same factor for every x ; normalization over admitted vectors is uniform. The cap affects only resolution probability.

5. Completion, retry, and persistence evidence

Evidence shall distinguish rejected candidates, admitted-member failure, scientific replacement, and idempotent execution retry. All six counts shall reconcile; every admitted member shall complete before UNC admission. A retry retaining scientific identity shall prove identical work and distinct I_attempt. Persistence evidence shall prove the selected mode's exact reconstruction, regeneration, sufficiency, lost questions, lifecycle, completion, and failure semantics.

6. UNC evidence

Evidence shall reproduce the equal-weight known-zero-center second moment on one exact common all-member domain without mean subtraction or B-1. It shall use the exact primary product name and separately state equality to target-law marginal variance. Counts, uniqueness, r_{sign} , r_{map} , effective information, and estimator uncertainty shall remain separate. Source and deterministic-residual imprint and every prohibited interpretation shall be visible.

7. Reciprocal and STD evidence

No reciprocal ordinary-base evidence is presently evaluable. A future route requires exact owner disposition, finite-positive parent domain, separate method/profile/use, and no role promotion.

STD evidence shall bind independently realized `m_MAP` and compatible `sigma_cond`, reproduce `zeta_cond`, verify unit 1, and emit only the bounded claim. It shall show no interpolation, extension, implicit reciprocal, JINC substitution, or significance language.

8. Profiles, transforms, and review result

Evidence shall preserve request, applicability, eligibility, and realization as separate fields. Existing r0.18 profiles and Registry objects remain immutable. Changed r0.3 evaluation is unavailable until exact approved/bound successors exist. External transform/Wiener/FRUIT evidence shall bind exact external authority and new generations for any learning or replay.

REQUIREMENT_PREDICTION_TRACEABILITY.csv supplies one row per stable requirement with owner decision, exact source digest, dependency, and parity. A future review reports pass, fail, unavailable, or not applicable for every row. This draft supplies no current conformance, validation, performance, readiness, freeze, or production result.

Appendix A. Exact Requirement/Prediction/Owner Traceability

SCI-NOI-REQ-001

Predictions: SCI-NOI-PRED-011 Owner decisions: r0.3-architecture-basis Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: SCOPE|TAX Parity: exact ECS section/review: 1 / operation-role review

SCI-NOI-REQ-002

Predictions: SCI-NOI-PRED-004 Owner decisions: ODQ-102A Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-identity-numerically-unavailable Dependency: PTCB|MAPB Parity: exact ECS section/review: 2 / operator identity review

SCI-NOI-REQ-003

Predictions: SCI-NOI-PRED-010 Owner decisions: ODQ-101|ODQ-104 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: GRAPH|FILTER Parity: exact ECS section/review: 8 / fixed-relearned review

SCI-NOI-REQ-004

Predictions: SCI-NOI-PRED-002 Owner decisions: ODQ-102B Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: PTCB|DESIGN Parity: exact ECS section/review: 3 / coherence audit

SCI-NOI-REQ-005

Predictions: SCI-NOI-PRED-001 Owner decisions: ODQ-102C Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: DESIGN|MAPB Parity: exact ECS section/review: 3 / mass-balance audit

SCI-NOI-REQ-006

Predictions: SCI-NOI-PRED-015 Owner decisions: SCI-NOI-OWNER-Q-SCOPE-01 Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: SCOPE Parity: exact ECS section/review: 1 / observation-array scope audit

SCI-NOI-REQ-007

Predictions: SCI-NOI-PRED-004 Owner decisions: ODQ-102A Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-identity-numerically-unavailable Dependency: PTCB|MAPB Parity: exact ECS section/review: 2 / sign-placement audit

SCI-NOI-REQ-008

Predictions: SCI-NOI-PRED-004 Owner decisions: ODQ-102A Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: MAPB Parity: exact ECS section/review: 2 / representation and parent audit

SCI-NOI-REQ-009

Predictions: SCI-NOI-PRED-012 Owner decisions: ODQ-102A Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: MAPB|GRAPH Parity: exact ECS section/review: 2 / response parity review

SCI-NOI-REQ-010

Predictions: SCI-NOI-PRED-015|SCI-NOI-PRED-016 Owner decisions: ODQ-102D|SCI-NOI-OWNER-Q-102D-01 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b|4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: DESIGN Parity: exact ECS section/review: 4 / population-state audit

SCI-NOI-REQ-011

Predictions: SCI-NOI-PRED-015|SCI-NOI-PRED-016 Owner decisions: SCI-NOI-OWNER-Q-102D-01 Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: DESIGN Parity: exact ECS section/review: 4 / edge-state audit

SCI-NOI-REQ-012

Predictions: SCI-NOI-PRED-014 Owner decisions: SCI-NOI-OWNER-Q-102D-01 Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: DESIGN Parity: exact ECS section/review: 4 / rejection mechanics audit

SCI-NOI-REQ-013

Predictions: SCI-NOI-PRED-013 Owner decisions: r0.3-architecture-basis|SCI-NOI-OWNER-Q-102D-01 Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: DESIGN|LIFE Parity: exact ECS section/review: 3 / universal design identity review

SCI-NOI-REQ-014

Predictions: SCI-NOI-PRED-014 Owner decisions: SCI-NOI-OWNER-Q-102D-01 Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: DESIGN Parity: exact ECS section/review: 4 / conditional-uniformity proof

SCI-NOI-REQ-015

Predictions: SCI-NOI-PRED-005 Owner decisions: SCI-NOI-OWNER-Q-102D-01 Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: DESIGN Parity: exact-repaired ECS section/review: 4 / duplicate-complement audit

SCI-NOI-REQ-016

Predictions: SCI-NOI-PRED-005 Owner decisions: SCI-NOI-OWNER-Q-102D-01|ODQ-105B Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5|272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: DESIGN|UNCT Parity: exact ECS section/review: 5 / count-rank-information review

SCI-NOI-REQ-017

Predictions: SCI-NOI-PRED-003|SCI-NOI-PRED-007 Owner decisions: ODQ-103 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: DESIGN Parity: exact ECS section/review: 6 / source-imprint claim review

SCI-NOI-REQ-018

Predictions: SCI-NOI-PRED-008 Owner decisions: ODQ-105A Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: LIFE|VALP Parity: exact ECS section/review: 5 / atomic completion audit

SCI-NOI-REQ-019

Predictions: SCI-NOI-PRED-013 Owner decisions: r0.3-architecture-basis Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: LIFE Parity: exact ECS section/review: 5 / retry-replacement identity audit

SCI-NOI-REQ-020

Predictions: SCI-NOI-PRED-009 Owner decisions: ODQ-109 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: LIFE Parity: exact ECS section/review: 5 / persistence plan review

SCI-NOI-REQ-021

Predictions: SCI-NOI-PRED-010 Owner decisions: ODQ-110A Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-boundary-numerically-unavailable Dependency: FILTER Parity: exact ECS section/review: 8 / external transform parity review

SCI-NOI-REQ-022

Predictions: SCI-NOI-PRED-010 Owner decisions: ODQ-110B|ODQ-110C Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-boundary-numerically-unavailable Dependency: FILTER|LIFE Parity: exact ECS section/review: 8 / generation separation review

SCI-NOI-REQ-023

Predictions: SCI-NOI-PRED-006|SCI-NOI-PRED-007 Owner decisions: ODQ-105B|ODQ-106 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-estimand Dependency: UNCT Parity: exact ECS section/review: 6 / UNC completeness review

SCI-NOI-REQ-024

Predictions: SCI-NOI-PRED-008 Owner decisions: ODQ-105A|ODQ-111 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-r0.18-profile-r0.3-evaluation-unavailable Dependency: UNCT|VALP Parity: exact ECS section/review: 6 / all-member admission audit

SCI-NOI-REQ-025

Predictions: SCI-NOI-PRED-002|SCI-NOI-PRED-006 Owner decisions: ODQ-105B Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-estimand Dependency: UNCT Parity: exact ECS section/review: 6 / estimator reproduction

SCI-NOI-REQ-026

Predictions: SCI-NOI-PRED-007|SCI-NOI-PRED-009 Owner decisions: ODQ-105B|r0.3-architecture-basis Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b|4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: UNCT Parity: exact ECS section/review: 6 / product-name-unit-claim review

SCI-NOI-REQ-027

Predictions: SCI-NOI-PRED-009 Owner decisions: ODQ-105B Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: UNCT Parity: exact ECS section/review: 6 / uncertainty-information separation

SCI-NOI-REQ-028

Predictions: SCI-NOI-PRED-005 Owner decisions: ODQ-106 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: UNCT|TAX Parity: exact ECS section/review: 6 / representation-role review

SCI-NOI-REQ-029

Predictions: SCI-NOI-PRED-006 Owner decisions: ODQ-107|r0.3-no-new-reciprocal-decision Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b|4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: legacy-approved-r0.3-successor-unavailable Dependency: UNCT|VALP Parity: typed-unavailability ECS section/review: 7 / reciprocal gate review

SCI-NOI-REQ-030

Predictions: SCI-NOI-PRED-017 Owner decisions: ODQ-108|ODQ-102A Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-method-numerically-unavailable Dependency: STDT|MAPB|JINCB Parity: exact ECS section/review: 7 / numerator-scale-parent audit

SCI-NOI-REQ-031

Predictions: SCI-NOI-PRED-017 Owner decisions: ODQ-108 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: STDT Parity: exact ECS section/review: 7 / unit-claim review

SCI-NOI-REQ-032

Predictions: SCI-NOI-PRED-011 Owner decisions: ODQ-111 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-r0.18-bytes-r0.3-successor-unavailable Dependency: VALP|REG_BIND Parity: exact-boundary ECS section/review: 8 / profile projection review

SCI-NOI-REQ-033

Predictions: SCI-NOI-PRED-011 Owner decisions: ODQ-111 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-r0.18-bytes-r0.3-successor-unavailable Dependency: VALP|REG_BIND Parity: exact-boundary ECS section/review: 8 / GEN-input profile review

SCI-NOI-REQ-034

Predictions: SCI-NOI-PRED-008|SCI-NOI-PRED-011 Owner decisions: ODQ-111 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-r0.18-bytes-r0.3-successor-unavailable Dependency: VALP|REG_BIND Parity: exact-boundary ECS section/review: 8 / UNC-member profile review

SCI-NOI-REQ-035

Predictions: SCI-NOI-PRED-008|SCI-NOI-PRED-011 Owner decisions: ODQ-111 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved-r0.18-bytes-r0.3-successor-unavailable Dependency: VALP|REG_BIND Parity: exact-boundary ECS section/review: 8 / UNC-ensemble profile review

SCI-NOI-REQ-036

Predictions: SCI-NOI-PRED-011|SCI-NOI-PRED-017 Owner decisions: ODQ-111|ODQ-108 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: r0.18-approved-r0.3-successor-unavailable Dependency: VALP|STDT|REG_BIND Parity: exact-boundary ECS section/review: 8 / STD successor gate review

SCI-NOI-REQ-037

Predictions: SCI-NOI-PRED-012 Owner decisions: r0.3-architecture-basis Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: CONV Parity: exact ECS section/review: 1 / typed-state no-default review

SCI-NOI-REQ-038

Predictions: SCI-NOI-PRED-010 Owner decisions: ODQ-110A|ODQ-110B|ODQ-110C Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: FILTER|LIFE Parity: exact ECS section/review: 8 / immutable ancestry review

SCI-NOI-REQ-039

Predictions: SCI-NOI-PRED-009 Owner decisions: ODQ-102C|ODQ-105B|ODQ-107 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: approved Dependency: TAX|UNCT|STDT Parity: exact ECS section/review: 7 / role-promotion prohibition

SCI-NOI-REQ-040

Predictions: SCI-NOI-PRED-011 Owner decisions: ODQ-111 Owner digest: 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b Approval: r0.18-approved-and-bound-r0.3-successor-unavailable Dependency: VALP|REG_BIND Parity: exact-boundary ECS section/review: 8 / profile immutability review

SCI-NOI-REQ-041

Predictions: SCI-NOI-PRED-001 Owner decisions: r0.3-architecture-basis Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: COVER Parity: exact ECS section/review: 8 / module binding review

SCI-NOI-REQ-042

Predictions: SCI-NOI-PRED-017 Owner decisions: r0.3-architecture-basis Owner digest: 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5 Approval: approved Dependency: SCOPE|COVER Parity: exact ECS section/review: 8 / claim-ceiling review